Achintya Rai

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Education

University of California, San Diego — Sept. 2024 – June 2028
Bachelor of Science in Computer Engineering — ***GPA: 3.9/4.0***

Work Experience

Adobe — San Jose, CA
Software Engineering Intern — *June 2026 – Sept. 2026*

- Profiled **60.6M+** payment events in Databricks, flagging **86,387** duplicate records and a **47%** timestamp null rate to establish the authoritative data source for downstream monitoring.
- Diagnosed a **26-hour** outage across **1,272** segments as the validation benchmark.
- Designed a time-series anomaly detector benchmarking algorithms across **30+** countries.
- Built SQL and MLOps pipelines automating weekly retraining and hourly inference at scale.

VALT Health — Remote
Founding Software Engineer — *Aug. 2025 – Feb. 2026*

- Engineered a Flask backend to process **100k+** datapoints from fitness devices (Fitbit, Apple Watch), designing database tables and schemas to unify data formats across **20+** distinct sources.
- Implemented async batch processing in GCP, cutting redundant DB writes **60%**.
- Automated environment setup and data seeding via Bash scripts, cutting testing time up to **80%**.

Decklar (formerly Roambee) — Sunnyvale, CA
Software Engineering Intern — *June 2024 – Aug. 2024*

- Developed real-time shipment monitoring modules for **Amazon, Coca-Cola, and Samsung**, optimizing Python and NumPy pipelines for a **15%** capacity boost across **1M+** weekly events.
- Implemented advanced data structures for analytics, yielding **20%** faster reporting.
- Refined a Panels-based UI through iterative user testing, improving workflow efficiency by **15%**.

Research

Graph Theory & Neural Architecture — San Diego, CA
Undergraduate Research Fellow (NSF REU) — *March 2026 – Present*

- Engineered a Python pipeline analyzing LLM embedding manifolds (Llama-2, DeepSeek-V2), reducing **500GB** checkpoints to **13GB** localized manifolds via weight-matrix extraction utilities.
- Optimized preprocessing for a **30x runtime reduction** on **8GB RAM** systems.
- Applied Rent's Rule and Louvain clustering to raise ICN density to **13.25%**.

Applications of Multi-Agent Systems — San Diego, CA
Undergraduate Research Fellow (UC Scholars) — *June 2025 – Aug. 2025*
Multi-agent AI Chess Tutor | *Django, React, MCP, SQLite*

- Built an AI chess analysis tool with **Django (Python)**, **React (TypeScript)**, and **SQLite**.
- Architected multi-agent **MCP** workflows, cutting hallucinations **80%**.
- Developed **7 REST API endpoints** and prototypes validated by **15+** chess players and coaches.

Projects

SquaredUpChess | *React, TypeScript, Django, WebSockets, AWS, Docker* — Aug. 2025 – Dec. 2025

- Engineered a full-stack platform on **AWS** with **20+ Django REST endpoints**.
- Built a responsive **React** UI and a **CI/CD** pipeline with automated testing and error logging.

MovieRecapinator | *Java, Spring Boot, Maven, JavaScript, IMDb API* — April 2025

- Built a **Spring Boot** recommendation engine in **Java** processing **700,000+** IMDb entries.

NanoGPT | *Python, PyTorch, Transformers, Neural Networks* — Mar. 2026

- Implemented a transformer in **Python**/**PyTorch** with multi-headed attention.

Technical Skills

Languages: Python, Java, C/C++, SQL (Postgres/SQLite), TypeScript, JavaScript, Bash, HTML/CSS
Development: React, Django, Node.js, FastAPI, Flask, Asyncio, WebSockets, JUnit, PyTest
AI & Cloud: PyTorch, TensorFlow, AWS, GCP, Docker, GitHub Actions, Databricks, MCP, Linux